

Said in Public, Lost in Recall: Why LLM Agent Societies Need Provenance-Aware State Updates

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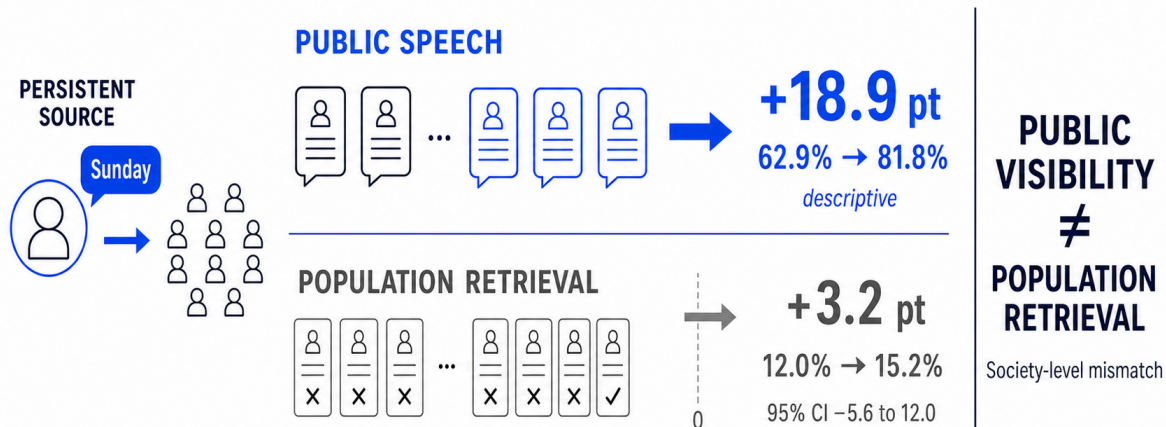


Figure 1: Speech is not retrieval. A persistent source changes the pooled current-mention share by +18.9 points but private current answers by only +3.2 points (95% CI -5.6 to 12.0), revealing a society-level mismatch.

Abstract

When a fact changes in an LLM agent society, the transcript can look corrected before the society has actually updated. We expose this failure in a 25-agent telephone-style task that follows an event update from received evidence (HEARD), through public utterances (SAID), to private retrieval (HELD). Keeping an authoritative source active raises the pooled current-mention share among value-bearing utterances from 62.9% to 81.8% (+18.9 points, descriptive), yet current private answers move only from 12.0% to 15.2% (+3.2 points, 95% CI -5.6 to 12.0). Authority therefore changes what is publicly visible without producing a supported gain in what the population can retrieve.

Where does the update disappear? We freeze every listener’s received evidence and replay the same ordered streams through alternative memories. Provenance-aware integration raises current answers from 32.0% to 60.5% over a Generative-Agents-style pipeline (+28.5 points, 95% CI 21.5–35.5), and retains a +26.0-point gain on independently generated streams supplied with controlled-task versions. In a stricter matched ablation, changing only conflict selection from mention frequency to maximum version adds 9.0 points (95% CI 4.4–13.6). These interventions identify listener-side integration as one causal locus and support a simple design principle: mutable facts should be treated as ordered state changes, not popularity contests among repeated claims.

The repair, however, reveals a second bottleneck. Default dialogue preserves no parseable source/version marker in 720 utterances, and unauthenticated versions can be forged. Reliable agent

societies therefore require more than a better memory module: they need a provenance-aware state-update protocol that preserves, resolves, and authenticates updates between communication and memory—alongside evaluation that tests what agents can retrieve rather than only what their transcripts display.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; *Natural language generation*.

Keywords

LLM agents, multi-agent systems, social simulation, memory, provenance, misinformation

ACM Reference Format:

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1 Introduction

LLM-based agent societies increasingly coordinate through conversation: they simulate communities, negotiate plans, deliberate, and divide work [24, 38, 49, 57]. Their success is therefore often read from the conversation itself. If the corrected fact appears repeatedly in a public transcript, the society seems corrected. But the transcript is a public record, not the retrievable state of the agents that must

later act. A fact may win the conversation without updating the society.

This distinction becomes critical when facts change. Standard conversational memories rank observations by recurrence, relevance, or salience. Those signals are useful for recalling a topic, but a mutable fact is not a majority vote: ten repetitions of an obsolete value should not override one authenticated update. Reliable coordination therefore requires more than transmission. It requires a rule that turns received claims into an ordered, retrievable state. Existing transcript-level evaluations blur these operations and cannot show where an update is lost.

We make the loss observable with a minimal changing-fact probe. A society of 25 agents initially knows an obsolete event time and location; one source then receives the current value, and agents propagate it through pairwise conversation. We follow the update across three operational stages: HEARD, evidence received by an agent; SAID, claims present in public utterances; and HELD, the answer returned in a private post-transmission interview. “Held” denotes an elicited behavior, not a literal mental state. The separation reveals the paper’s central observation: an authoritative source can make the update more visible in public speech without a supported increase in population-wide private retrieval.

We then ask a narrower question than whether a model can state the truth: given exactly the same received evidence, which memory operation determines what a listener later returns? Matched-event replays freeze each agent’s realized stream before comparing complete listener-side memories. A stricter ablation holds representation, candidates, retrieval, context, and interview fixed while changing only conflict selection from mention frequency to maximum version. Together these interventions locate one failure inside listener integration and test provenance-aware updating as a repair. Follow-up channel and attack tests establish the repair’s boundary: provenance must survive relay and be authenticated before a version can safely govern state.

This paper makes three contributions:

- (1) an empirical finding: public uptake and population retrieval can diverge, which makes transcript-only success an incomplete measure of social coordination;
- (2) a causal diagnosis: fixed-stream interventions locate part of the failure in listener-side integration and isolate the benefit of version priority over repetition; and
- (3) a protocol-level design principle: mutable facts require a communication–memory interface that preserves and authenticates provenance before applying ordered state updates.

2 Related Work

LLM agent societies and communication. Generative Agents established memory, reflection, and retrieval as a basis for believable long-horizon social behavior [38]; CAMEL, AutoGen, and SOTOPIA developed communicative multi-agent interaction and social evaluation [7, 24, 49, 57]. Population-scale simulation and role-specialized frameworks broaden this setting [16, 18, 39, 40], while theory-of-mind reasoning, communication-graph design, and learned message representations target coordination more directly [25, 41, 55]. Debate-oriented systems study whether interaction improves task

accuracy [13, 21, 27, 34], although protocol comparisons show that gains are sensitive to design choices [43]. Failure taxonomies and adversarial studies further expose specification, verification, and compounding error problems [2, 6, 17, 19, 50]. We study decentralized propagation of a changed fact rather than a centralized final decision or a generic coordination failure.

Public expression and private agent state. Several adjacent studies show why transcripts alone are insufficient. Persona simulations compare pre- and post-discussion private responses with chat transcripts and find conformity and instability [3, 48]. Think-Before-Speak explicitly separates structured internal evaluation from public expression in social simulation [52]. Embodied-agent work measures alignment between private world models and finds that dialogue can reduce action conflict while degrading task success [11]. Our distinction is operational and fact-specific: HEARD records received events, SAID measures claims in public utterances, and HELD probes the answer retrieved after transmission. We then intervene on the listener’s memory rule rather than only describing a public/private gap.

Misinformation, conformity, and factuality. Human misinformation research distinguishes exposure and diffusion from durable correction [22, 23, 45, 46]. Factuality benchmarks and detectors likewise treat semantic truth as distinct from surface plausibility [14, 26, 29, 30, 36]. Rumor cascades, LLM-mediated misinformation, and audience-tuned language show that transmission can transform both content and expression [4, 15, 31, 54], whereas epidemic and randomized rumor-spreading algorithms characterize reach and latency under explicit message semantics [10, 20]. Our current/stale/unknown verdict is analogous to three-way fact verification [44], but is applied to post-communication agent answers. Attacks on multi-agent collaboration and concurrent work on error cascades examine how false inputs spread and how centralized governance may suppress them [2, 50]. Our complementary setting starts from a valid correction to a formerly true fact, probes each listener privately, and evaluates a local memory interface.

Agent memory and belief revision. Long-term agent memories improve storage, retrieval, and organization [8, 37, 51, 56]; probabilistic belief memory retains competing uncertain conclusions under partial observability [28]. Reflection and acting systems use textual memory to improve sequential decisions [42, 47, 53], while factual editing changes parameters rather than an agent’s external memory interface [32, 33]. We compare controlled motif adaptations under a common harness, not the official systems end to end. Provenance itself is not new: truth-maintenance and belief revision formalize how new information supersedes prior commitments [1, 12], while data provenance records origin and derivation [5, 35] and opinion dynamics studies source-weighted updating [9]. Our contribution is an empirical bridge between these lines: under frozen social evidence, supplied versions affect listener integration, but the benefit is bounded by whether the communication channel preserves and authenticates them.

3 Following an Update from Speech to State

3.1 A changing-fact telephone game

The central story requires separating three questions that a transcript normally conflates: did an agent receive the update, did the update appear in public speech, and could the agent retrieve it afterward? We operationalize those questions in the harness behind Figure 2. A society contains $N = 25$ agents connected by a contact graph. The agents initially discuss a stale event value v_0 (Saturday at the front porch). At round 1, the environment gives one designated source the current value v_1 (Sunday at the community center). During each round, scheduled pairs exchange natural-language utterances. Each agent observes both sides of its encounters, consolidates those observations into memory, and retrieves memory to condition later speech. After the final round, a separate interview asks each agent when and where the event is now held.

Each scenario registers surface markers for its current and stale event values. The deterministic interview scorer assigns *current* when an answer contains at least one current marker and no stale marker, *stale* whenever a stale marker appears (including a mixed answer), and *unknown* otherwise. The three-way split distinguishes omission from active return of superseded content. SAID measures current mentions among final-round utterances that contain at least one registered value marker; current and stale mention flags are non-exclusive for mixed utterances. HELD is the current share among all private interviews. HEARD denotes the ordered evidence record delivered to a listener; reach and evidence composition are derived from that record without assuming that receipt implies adoption.

The primary verdict is a deterministic marker classifier parameterized by the registered current and stale values for each scenario. We audited all 375 interview answers in the repair-drive baseline, source, and broadcast experiment with a separate semantic LLM judge. Agreement was 374/375 overall: 124/125 baseline answers and all 250 source and broadcast answers. Both scorers produced the same current counts in every condition. This validates the headline interview metric on the core sample; it does not validate every auxiliary scenario or every free-form claim in the transcripts.

3.2 Formalization

Let q index an experimental condition, $r \in \{1, \dots, K_q\}$ a schedule seed, and $i \in \{1, \dots, N\}$ an agent. Let U_{qrT} be the final-round public utterances and $h_{qri} \in \{c, s, \emptyset\}$ the verdict on agent i 's private interview answer. For any text x , $C(x)$ and $S(x)$ indicate a registered current or stale marker, and $V(x) = C(x) \vee S(x)$ indicates a value-bearing text. The two society-level readouts are

$$\text{Said}_{qr} = \frac{\sum_{u \in U_{qrT}} \# [C(u)]}{\sum_{u \in U_{qrT}} \# [V(u)]},$$

$$\text{Held}_{qr} = \frac{1}{N} \sum_{i=1}^N \# [h_{qri} = c].$$

The headline SAID statistic pools its numerator and denominator across seeds and is descriptive; HELD is averaged within each seed, with seeds as the independent units for intervals and paired tests. Mixed utterances may satisfy both C and S , but enter the SAID denominator once; unknown and stale interviews remain in the

HELD denominator. The full ordered evidence record is HEARD; Appendix C.1 defines its reach and composition summaries.

The central dissociation is a contrast between conditions, not the numerical subtraction of SAID and HELD:

$$\Delta_{\text{Said}} = \text{Said}_{\text{source}}^{\text{pool}} - \text{Said}_{\text{baseline}}^{\text{pool}},$$

$$\Delta_{\text{Held}} = \frac{1}{K} \sum_{r=1}^K (\text{Held}_{\text{source},r} - \text{Held}_{\text{baseline},r}).$$

Empirically, the pooled descriptive $\Delta_{\text{Said}} = 18.9$ percentage points, whereas $\Delta_{\text{Held}} = 3.2$ points (95% CI -5.6 to 12.0). The two effects use different observational units and establish a society-level response dissociation, not an agent-level joint contingency.

3.3 Conditions and experimental units

The core experiments use five communication rounds and two pairwise meetings per agent per round. Each meeting contains three alternating short utterances, and both participants observe every utterance in that meeting. A schedule seed fixes the random pairings; seeds 41–48 are reused in paired comparisons. The source receives the changed fact after round-one encounters, so its first opportunity to relay the version is round two. The baseline is a Generative-Agents-style (GA) stream memory with retrieval and consolidation. We vary one factor at a time: model capability, meeting density, memory organization, persona detail, a persistent authoritative source, and an all-agent broadcast. The broadcast writes the current value to every agent and is therefore an overwrite positive control, not a decentralized repair.

Unless noted otherwise, a schedule seed defines one independently generated society and is the statistical unit. We report percentages aggregated over agents, seed-level 95% t intervals where available, and paired tests when conditions share seeds. The central architecture and capability comparisons use eight seeds (200 agent interviews per condition). Smaller APM adversarial runs are identified as pilots rather than pooled with the main evidence.

3.4 Models, prompts, and execution

The main model string is gpt-5.4-mini. The model-family check uses DeepSeek-V4-Flash. Calls use a third-party OpenAI-compatible Responses endpoint; core societies were collected on June 19–21, 2026. The nominal temperature is zero, although the access configuration omits that field when unsupported by the endpoint. The client sets no explicit context-window override. The conversation system prompt asks for one short in-character utterance grounded in the retrieved memory notes. Its user message contains the speaker persona, partner, topic, retrieved notes, and the utterances already produced in that meeting. The interview uses a separate prompt: answer the question using only the memory notes and return one short JSON sentence. Interview calls are read-only and cannot alter memory or later communication.

All model calls are required to return the requested JSON field. Transport failures are retried; an incomplete agent condition is not converted to *unknown* and is rerun from per-agent checkpoints. Raw round logs, final memory snapshots, interview JSON, schedule seeds, reconstructed run configurations, aggregation scripts, and the exact prompt strings are included in the anonymous artifact. The oldest run configurations were reconstructed from immutable

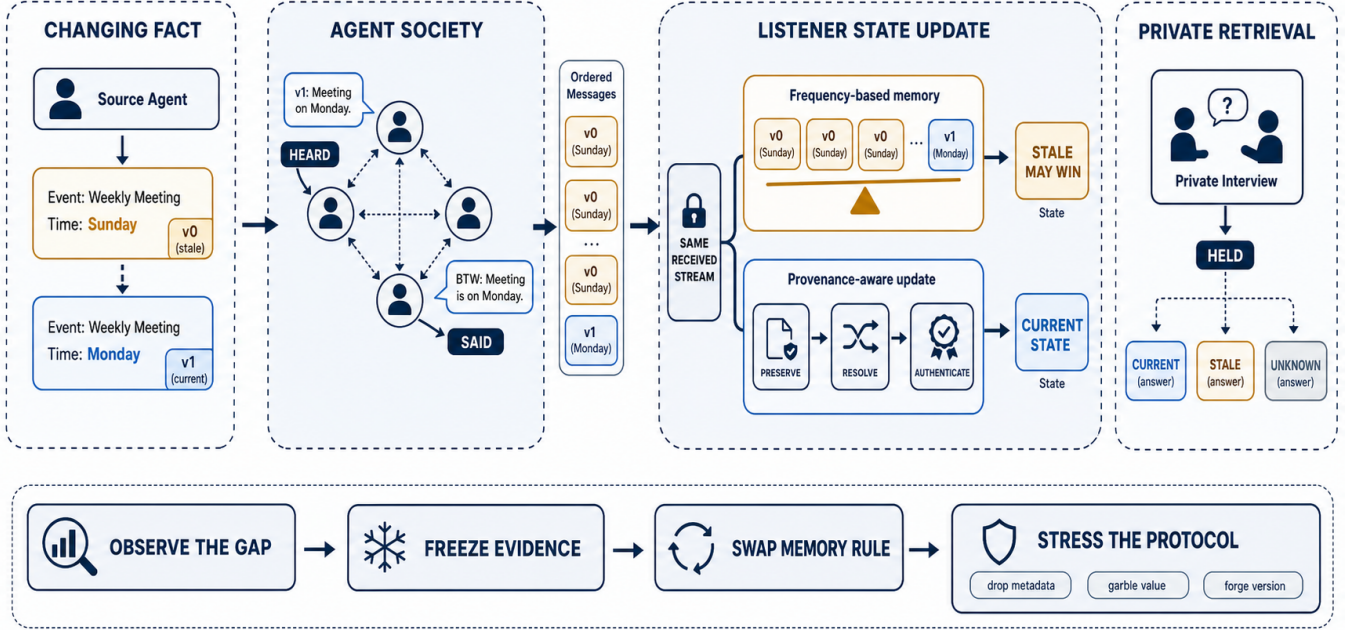


Figure 2: From speech to state. We observe **HEARD**, **SAID**, and **HELD**, then freeze each received stream and replace only the listener’s update rule. Channel and attack tests stress preservation, resolution, and authentication.

aggregate rows and the experiment ledger; those files are explicitly marked as reconstructed.

3.5 Memory pipelines and controlled adaptations

The GA baseline appends every observed event. At each round boundary it asks the model for up to three high-level reflections over the ten most recent events. Retrieval ranks all stored event texts against the query and returns six events plus three reflections. Semantic embedding cosine is attempted when the optional local model is available, otherwise retrieval falls back to a deterministic lexical ranker. Historical runs did not persist the backend selected on each call; the artifact records this reproducibility boundary. Consequently, current and stale claims can appear together in the interview context, and reflection may add another free-text interpretation.

The PROV pipeline also stores the raw stream, but additionally maintains a compact pair (z_i, v_i) . An incoming typed event changes that pair only if its version is larger. Retrieval prepends the selected pair as “latest update, version z_i ” and then returns four relevance-ranked raw events; consolidation is a no-op. Thus, PROV differs from GA in typed representation, online conflict resolution, absence of free-text reflection, and explicit retrieval salience. We retain this complete system comparison because it measures an implementable interface, while the selector-only ablation identifies one component of its gain.

The architecture figure uses controlled adaptations of external memory motifs. Raw returns recent relevant events without reflection; the current-focused GA variant changes only the retrieval query; the Mem0 motif extracts and updates a compact fact; the A-MEM motif evolves linked notes; and the MemoryBank motif adds recency and forgetting behavior. Each adaptation uses the same society, prompt, model, and interview. This common harness supports a mechanism comparison but omits system-specific orchestration, training, and storage layers; these labels are therefore not official benchmark scores for the cited projects.

3.6 Provenance-aware integration

The Prov memory associates an observed claim with a version z , value v , and immediate source. For listener i with current memory state (z_i, v_i) , observation $e = (z_e, v_e, s_e)$ updates state by

$$(z_i, v_i) \leftarrow \begin{cases} (z_e, v_e), & z_e > z_i, \\ (z_i, v_i), & \text{otherwise.} \end{cases}$$

An agent relays the version attached to the state it currently retrieves. The rule is local: agents that never receive the update are not overwritten. It differs from the baseline in how received evidence is represented, consolidated, retrieved, and used for the final answer.

The protocol assumes that source and version survive relay as structured metadata. This is an idealized systems intervention, comparable to adding typed fields to an agent communication interface. We separately test text-only dialogue and find that such fields

are not reliably recoverable unless an explicit attribution norm is imposed. Accordingly, our claim is conditional on provenance-preserving communication.

3.7 Matched-event replay

In an ordinary end-to-end comparison, memory affects later speech; GA and PROV may therefore receive different downstream evidence. To isolate listener handling, we take eight realized PROV societies and reconstruct every per-agent observation in its original order. Each event contains the utterance text, round, speaker, listener, and the provenance fields present to the original system. We verify reconstruction by matching every agent’s recovered final version and value to its saved PROV state.

For each seed, the same frozen per-agent stream is then replayed into fresh GA and PROV memories. Round boundaries, interview question, model, and scoring are shared, but each memory retains its own representation, consolidation, retrieval, and context construction. No replayed answer can affect another agent’s stream. Provenance reaches 10–19 of 25 agents depending on the seed, so the replay is not an all-agent overwrite. Because the source streams were realized under PROV, this test estimates a complete listener-side memory-pipeline contrast on that support; it neither isolates one rule nor estimates a fully architecture-neutral conversation distribution.

For memory pipeline a , let $\text{Held}_r^{(a)}(E_r)$ be its current-answer rate under the frozen stream set E_r . Here $E_r = \{E_{ri}\}_{i=1}^N$ contains one ordered stream per agent. The complete-pipeline repair estimand is

$$\Delta_{\text{pipe}} = \frac{1}{K} \sum_{r=1}^K \left[\text{Held}_r^{(\text{Prov})}(E_r) - \text{Held}_r^{(\text{Ga})}(E_r) \right].$$

This contrast localizes one causal contribution to listener-side handling on the support of PROV-generated evidence; it does not imply that memory is the only cause of the dissociation.

3.8 Strict matched-policy ablation

To isolate the conflict rule, we construct an isomorphic replay on the same eight streams. Every recognizable event claim is normalized into one typed candidate schema with arrival index, value, and version. Existing structured versions are retained. Because the task registers the original Saturday/front-porch value as version 0 and the Sunday/community-center update as version 1, unambiguous text-only mentions can receive the corresponding analysis annotation. Text that mentions both values without structured provenance is excluded as ambiguous.

The two conditions share the normalized candidates, store, no-op consolidation, one-record retrieval budget, context template, interview prompt, model, and scorer. Mention-frequency selection chooses the most frequent candidate value and breaks ties by latest arrival; maximum-version selection chooses the largest version and uses the same tie break. Across 5,412 received events, normalization retains 494 structured-provenance candidates and 556 unambiguous text candidates, excludes 40 ambiguous mentions, and ignores 4,322 events that mention neither registered value. The text-derived version is an oracle task annotation: this ablation tests what a listener does with a supplied version, not whether ordinary dialogue can infer or authenticate one.

3.9 Statistical analysis

The schedule seed, not the agent or utterance, is the independent unit. For each condition we compute a rate within each seed and report the mean with a two-sided 95% Student t interval across seeds. Paired conditions reuse schedules and are summarized by the seed-level difference. For the two fixed-stream studies we additionally report the exact two-sided sign test over non-zero paired directions. No multiplicity correction is applied to the supporting generality table; those cells are explicitly treated as robustness checks rather than separate confirmatory claims. Percentages may therefore differ from ratios of pooled counts when seed sizes differ; all core cells use 25 agents per seed.

4 Results

4.1 Public speech moves toward the update; population retrieval barely moves

The transcript first tells a convincing success story. Figure 1 shows that keeping the authoritative source active makes the current value substantially more prominent whenever the event is discussed. Population-wide interviews tell a different story. Figure 5 in the appendix reports the paired seed trajectories and the broadcast control. Under the baseline, 22 of 35 final-round value-bearing utterances contain a current marker (62.9%), but only 15/125 private interviews are current (12%; seed-level 95% CI 3–21%). Keeping the authoritative source active shifts the current-mention share to 27/33 (81.8%), while current HELD is 19/125 (15%; 10–21%). The paired HELD difference is only +3.2 points (95% CI –5.6 to 12.0). The SAID denominator deliberately excludes the roughly 320 final-round utterances per condition that mention neither event value; the result is a shift in what agents say *when they discuss the event*, not a claim that most dialogue concerns it. As a seed-level sensitivity check, the paired source-minus-baseline SAID difference averages +7.8 points (95% CI –40.3 to 56.0); the small and variable value-bearing denominators make this estimate imprecise. At the agent level, 10/14 baseline and 12/19 source agent–seed instances that said a current value subsequently returned a current private answer. Thus the headline mismatch is between public-transcript visibility and population-wide retrieval, not evidence that most individual agents publicly state and then lose the update. Broadcasting directly to every memory yields a current marker in all 226 value-bearing utterances (three also contain a stale marker) and 124/125 current interviews (99.2%), confirming that the answer can be expressed when transmission is bypassed.

More capability and contact do not close the gap. Before changing the memory rule, we tested whether familiar scale and interaction levers provide a simpler repair.

Table 4 summarizes these supporting controls. Increasing capability gives a modest, non-monotonic lift (18% to 34% to 30%) rather than a cure. A powered meeting-density rerun is approximately neutral-to-negative; a previously observed extreme connectivity cell did not replicate and is excluded. Rich personas and a persistent source likewise show no supported improvement. These controls do not prove equivalence, but they rule out large repairs from simply scaling model capacity, contact, characterization, or continued access to authority in this harness. The remaining question is not

how much agents communicate, but what their memories do with the evidence they receive.

4.2 Freezing the evidence locates failure inside listener memory

The public-private mismatch alone does not identify where the update is lost. Exposure predicts success within a condition: agents that hear a larger share of current mentions are more likely to return the current answer. Across conditions, however, mean current heard-fractions are similar (0.64 baseline, 0.72 source, 0.73 broadcast) while HELD outcomes are opposite (12%, 15%, 99%). Received frequency therefore underdetermines the final answer.

Matched-event replay provides the causal contrast. On identical text-plus-metadata streams, GA reaches 32.0% current HELD answers (95% CI 28.0–36.0), whereas PROV reaches 60.5% (52.8–68.2). The paired gain is 28.5 percentage points (21.5–35.5). It is positive in all eight schedule seeds; the exact two-sided sign test gives $p = .0078$. Since both memories receive the same event sequence, architecture-induced differences in later conversation cannot explain this replay gap. The result identifies listener-side *memory handling* as causally consequential under structured provenance: changing what a listener does with the same evidence changes what it can later return. Because the pipelines also differ in representation, consolidation, retrieval, and context salience, this 28.5-point contrast does not by itself assign the gain to one rule (Figure 3b).

A complementary cross-source replay uses eight realized GA society streams. Because those transcripts contain no native metadata, the controlled task supplies version 1 to unambiguous current mentions and version 0 to unambiguous stale mentions; mixed claims remain unannotated. On these identical text-plus-annotation streams, GA reaches 15.0% current answers and PROV 41.0%, a paired gain of 26.0 points (95% CI 16.5–35.5; positive in 8/8 seeds; exact sign test $p = .0078$). The listener effect therefore persists beyond PROV-generated linguistic support, conditional on correct task versions; it does not validate automatic provenance extraction.

4.3 Version priority, not repetition, resolves one conflict mechanism

The strict matched-policy ablation makes that assignment at a smaller scale. With every other operation held isomorphic, mention-frequency selection yields 62.5% current answers (95% CI 52.7–72.3), while maximum-version selection yields 71.5% (64.3–78.8). The paired difference is +9.0 points (4.4–13.6), positive in all eight seeds (exact two-sided sign test $p = .0078$). The 400 common-prompt interviews exactly match the selected symbolic state. Thus, supplied version priority itself has a positive causal effect; the remainder of the larger pipeline gap may arise from representation, retrieval salience, consolidation, or their interactions (Figure 3c).

Two mechanisms. The complete-pipeline and selector-only interventions improve different failure modes. In the complete pipeline comparison, PROV produces 121 current, 10 stale, and 69 unknown answers over 200 agents; GA produces 64 current, 15 stale, and 121 unknown. The 57 additional current answers are therefore accompanied mainly by 52 fewer unknowns rather than by a large stale-to-current conversion. This pattern is consistent with explicit

retrieval salience: once a versioned update reaches an agent, PROV places the selected state at the top of the interview context instead of asking the model to resolve a mixed event/reflection set. The paired gain in the net current-minus-stale rate is 31.0 points (95% CI 25.1–36.9).

The selector-only ablation has a sharper signature. Both policies leave the same 23 agents unknown because neither changes reach or candidate extraction. Maximum-version selection converts 18 outcomes from stale to current relative to frequency, changing pooled counts from 125/52/23 to 143/34/23. Seed 46 supplies the largest contrast: five agents receive a numerical majority of stale mentions after at least one version-1 update, so frequency returns stale while maximum version returns current. The fact that unknown is invariant in this ablation helps rule out retrieval volume or interview compliance as the explanation for its 9-point gain. This is the paper’s narrow mechanistic result: for a received mutable fact, order in the version sequence can matter more than popularity in the transcript.

4.4 The repair persists across controlled settings

Under one social harness, raw transcript storage reaches 14%; controlled Mem0, A-MEM, and MemoryBank adaptations, GA, and a current-focused GA variant range from 18% to 25%. PROV reaches 57%. The comparison should be read as a mechanism test, not a leaderboard: all adaptations share our simulator, prompts, and interface rather than reproducing each external system end to end.

The advantage persists beyond the repair-drive wording. Across three scenarios, current HELD is PROV/GA = 57/22 for a repair drive, 69/47 for a book club, and 59/18 for a carpool. A numeric dues change gives 59% versus 28%. These scenario and fact-type checks use five seeds except the eight-seed repair-drive architecture cell; their intervals are reported in Table 1. At round 10, random, ring, and small-world point estimates are 93/22, 68/6, and 83/10. On a second model family, DeepSeek-V4-Flash, the eight-seed comparison is 64.0% PROV versus 15.5% GA with disjoint intervals. Five-seed official-API coverage of the remaining memory rows ranges from 13.6% to 39.2%; none closes the gap to PROV.

These checks broaden the observed pattern but do not turn the controlled probe into a universal claim. Scenario surfaces share the same change-and-propagate structure, topology cells are point-estimate robustness checks, and only two model families are included.

4.5 Provenance is the missing interface between speech and memory

The structured intervention identifies what a reliable interface can do, but deployment also requires provenance to reach the interface. We test this dependency with text-coupled memory at round 10. Under default LLM dialogue, only 16 of 75 agents return the current value; none return the stale value, and 59 abstain as unknown. More importantly, zero of 720 utterances preserves a source/version phrase, so the provenance parser has almost nothing to integrate. Under a strong attribution norm, 610 of 720 utterances contain source/version-like markers and all 75 agents return the current value. The norm uses natural-language text rather than hidden

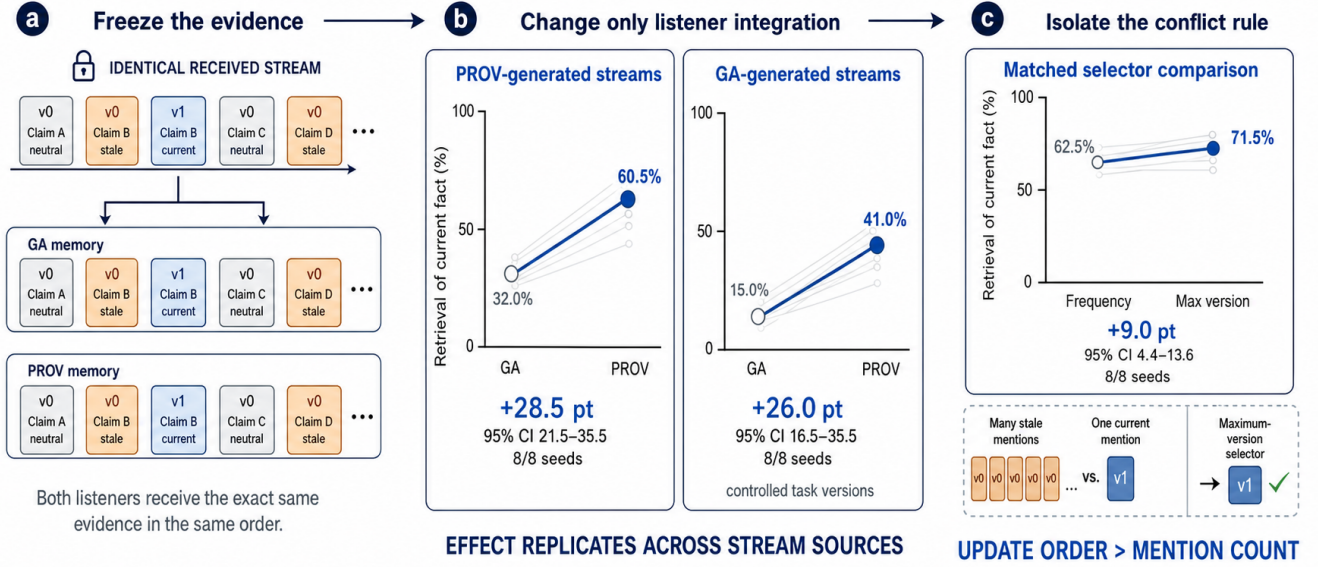


Figure 3: Same evidence, different memory. a, Both pipelines receive the same ordered stream. b, Complete Prov integration improves current HELD across both stream sources. c, With all other operations fixed, maximum-version selection outperforms mention frequency. All paired effects are positive in eight of eight seeds; intervals are shown in the panels.

Table 1: Memory comparison at a glance. Entries are current HELD (%); brackets are seed-level 95% intervals. Bold and underline mark the best and second-best multi-seed, non-ceiling results. Dashes are not scheduled or not applicable. Named systems are controlled motif adaptations, not official implementations.

Memory	Current HELD across tasks and settings							Summary	Update interface	
	Repair drive	Book club	Carpool	Numeric dues	Three-step update	Long horizon ($r = 10$)	DeepSeek V4-Flash	Mean $r = 5$ ↑ [§]	Explicit version	Relays source
Raw transcript	14 [6,23]	28 [0,56]	11 [0,31]	16 [0,35]	10 [1,20]	8 [0,30]	<u>39.2</u> [27.3,51.1]	17.4	no	no
Mem0 motif	18 [14,23]	<u>66</u> [29,100]	19 [10,28]	22 [8,37]	<u>19</u> [1,37]	30 [0,80]	<u>26.4</u> [16.1,36.7]	<u>31.4</u>	no	no
A-MEM motif	19 [12,26]	52 [38,66]	<u>30</u> [25,34]	25 [20,30]	14 [3,26]	18 [0,37]	<u>30.4</u> [12.3,48.5]	<u>31.4</u>	no	no
GA	22 [13,30]	47 [45,49]	18 [10,25]	<u>28</u> [18,38]	18 [8,27]	22 [15,28]	15.5 [10.1,20.9]	28.5	no	no
Current-focused GA	25 [15,35]	42 [17,67]	24 [17,31]	8 [1,15]	8 [1,15]	33 [13,53]	13.6 [5.3,21.9]	24.6	no	no
MemoryBank motif	<u>25</u> [19,31]	42 [4,81]	18 [4,33]	27 [19,35]	11 [3,19]	13 [0,28]	<u>26.4</u> [15.0,37.8]	28.2	no	no
Prov (ours)	57 [49,65]	69 [62,76]	59 [49,69]	59 [49,69]	68 [59,77]	93 [82,100]	64.0 [57.8,70.2]	61.0	yes	yes
Broadcast ceiling	99 [97,100]	–	–	–	–	–	–	–	n/a	n/a

Three-step uses three successive updates and three full post-final-update propagation rounds ($r = 7$); long-horizon uses $r = 10$. [§]Mean $r = 5$ is the equal-weight macro-average of Repair drive, Book club, Carpool, and Numeric dues; $r = 7$, $r = 10$, and DeepSeek cells are excluded. Long-horizon GA uses the eight-seed C17 reference and long-horizon Prov uses five seeds. DeepSeek GA and Prov use eight seeds; the other DeepSeek cells use five official-API seeds.

event fields, but its strength makes it a protocolized upper bound rather than ordinary conversation (Figure 4a).

These results separate two requirements that an end-to-end score would conflate: communication must preserve provenance, and the listener must integrate it. The matched-event replay establishes the second requirement conditional on the first; this text-only contrast shows why the first cannot be assumed. Provenance-aware memory

is therefore not a self-contained repair. It is one endpoint of a state-update interface spanning communication and memory.

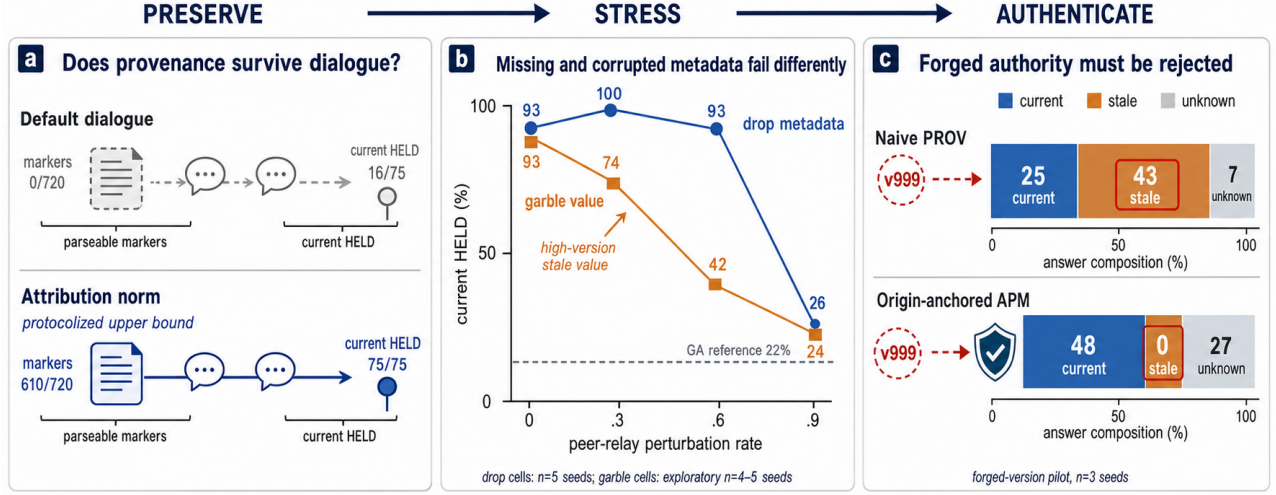


Figure 4: Provenance needs a protocol. a, Default dialogue preserves few usable markers; explicit attribution preserves them. b, Missing metadata degrades gradually, whereas corrupted high-version values mislead. c, Origin anchoring rejects the tested forged version that naive PROV accepts. Counts and sample sizes are shown in the panels.

4.6 Missing metadata degrades; corrupted metadata misleads

Structured metadata need not be perfectly transmitted to be useful, but corrupt metadata is more dangerous than missing metadata. We stress the round-10 repair-drive system with two independent per-relay perturbations. Under *drop*, the provenance payload is absent while the ordinary utterance is still delivered. Under *garble*, the payload keeps its high version but carries the stale value. The authoritative source’s direct receipt is never perturbed. Drop cells use five seeds; garble cells use four or five and are treated as exploratory point estimates.

Figure 4b shows redundancy under missing metadata: current answers remain 93–100% through 60% drop, then approach the GA reference at 90% drop. Garbling degrades more steadily because naive PROV treats a stale value with a high version as authoritative. At 60% garble it retains 42% current, but at 90% it is indistinguishable from the 22% GA reference at the precision of this exploratory check. These cells address the narrow objection that every relay must be lossless; they do not model correlated failure or an adaptive attacker.

The stress test also exposes a design defect. Once a garbled value is stored at version z , a later correct claim at the same version cannot replace it because the update rule requires $z_e > z_i$. A deployed protocol therefore needs same-version conflict handling rather than first-arrival stickiness: retain both candidates, validate the origin signature and content hash, or abstain until independent paths agree. The origin-anchored extension below addresses forged authority, but its small pilot does not exhaust these conflict cases.

4.7 Authentication turns provenance into a protocol

PROV adopts the highest version blindly. An adversary can therefore attach a forged version 999 to the stale value. We harden the protocol into Auditable Provenance Memory (APM) with three mechanisms: (1) an authorization flag minted only at the origin and relayed without escalation; (2) a source–version–path trace retained with every candidate; and (3) optional independent-path corroboration before commitment, with abstention represented as unknown. These mechanisms make each accepted update auditable and prevent an ordinary speaker from manufacturing origin authority.

In a three-seed, model-matched pilot, the forged stale claim leaves PROV at 33% current and raises stale answers to 57% (43 of 75 agents). Under the same attack, APM yields 64% current with no stale answer among 75 agents (Figure 4c). This is evidence for the specific defense against the specific attack, not a general adversarial guarantee. A separate sparse-communication check ($n = 8$, round 10) yields 40.5% current for APM and 21.5% for GA with overlapping current-answer intervals; the clearer safety difference is stale 0/200 for APM versus 30/200 for GA. We therefore treat APM as a promising hardening extension, not the paper’s primary efficacy claim.

5 Discussion

The transcript is not the society’s state. Public speech moved toward the update while population retrieval showed no supported gain, so a corrected-looking transcript did not establish a corrected society. Because SAID and HELD have different observational units, this is not evidence that most agents said the update and then forgot it. It is a gap between public visibility and population retrieval; evaluations should measure both rather than treating the first as a proxy for the second.

Mutable facts are state transitions, not votes. Fixed-stream interventions turn the observational gap into a diagnosis: changing

listener-side handling of the same evidence changed later answers. The matched selector isolated a 9.0-point benefit from maximum-version priority, supporting a specific principle: once a mutable fact reaches a listener, conflict resolution should respect update order rather than only repetition. This rule does not explain the entire 28.5-point pipeline gain; state salience, consolidation, and retrieval remain candidate mechanisms.

The repair is a protocol, not a memory module. Version-aware selection works only when communication supplies trustworthy versions. Default dialogue preserved no parseable source/version marker, and corrupted high-version metadata could install a stale value. A reliable interface must therefore connect reach, preservation, resolution, and authentication, with abstention when any stage fails (Appendix B). This systems layer—not provenance storage alone—is the broader implication of the repair.

Scope and alternatives. The experiments establish the value of correct provenance at one listener-side locus, not that integration is the only source of the gap or that deployed agents will infer trustworthy metadata automatically. The study covers 25-agent, short-horizon event tasks and few model families; HELD is elicited behavior, replays assume structured or controlled-task versions, and the APM pilot is not a security certification. Provenance interfaces also trade naturalism for auditability and may be unsuitable when provenance loss is itself under study.

6 Ethical and Social Considerations

The study uses synthetic personas and no human data. Provenance can improve auditability but may also enable surveillance or misplaced trust in credentials; deployment requires data minimization, credential protection, uncertainty disclosure, and stakeholder validation. Generative AI tools assisted coding, search, checks, and revision; the authors inspected the outputs.

7 Conclusion

A fact can win the conversation without updating the society. Separating HEARD, SAID, and HELD exposed that failure, while fixed-evidence replays located a consequential part of it in listener-side integration. Provenance-aware memory gained 28.5 points on identical streams, and the stricter selector intervention isolated a 9.0-point benefit from version priority over repetition. The resulting design lesson is bounded but concrete: mutable facts in LLM agent societies need a provenance-preserving, authenticated state-update interface between communication and memory, and evaluation must test retrievable state rather than infer it from public speech.

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A Reproducibility and Artifact Map

The anonymous artifact contains the simulator, memory code, prompts, replay scripts, aggregation scripts, normalized selector records, per-seed outputs, run configurations, a sanitized execution provenance record, and a data index. Core transcripts are included, while aggregate-only exploratory cells are labeled. Model identifiers and access-time configurations are recorded, but proprietary weights and provider infrastructure cannot be archived; fresh generations may therefore differ.

Three reproduction levels separate model-free checks from new generation. Headline numbers re-aggregate offline from accepted per-agent and per-seed JSON. The fixed-stream runner reconstructs eight societies, comprising 200 per-agent streams, and verifies every terminal version–value pair against the saved Prov snapshot before any model call. The selector runner reproduces normalization, outcomes, intervals, and exact sign tests offline; its online mode repeats the common-prompt interviews. Expected aggregates and a SHA-256 package manifest are included.

The schedule seed controls pairings and topology sampling but cannot make proprietary model generation bitwise deterministic. Saved accepted runs are therefore the evidential record, and each condition must contain all 25 agents. Reproduction proceeds by (1) verifying the manifest and environment lock; (2) re-aggregating accepted JSON; (3) reconstructing the eight societies and checking 200 terminal agent states; (4) rerunning selector normalization, intervals, and sign tests; and, optionally, (5) repeating interviews or full society generation. A missing agent, failed state check, or unregistered seed–configuration pair causes the corresponding condition to fail closed.

Table 2: Claim-to-artifact map.

Claim family	Required records
SAID/HELD gap	final-round logs, interviews, marker and semantic-judge audit
Complete replay	ordered observations, memory snapshots, replay checkpoints
Cross-source replay	GA streams, task annotations, paired checkpoints
Selector effect	normalized candidates, policy outputs, paired seed summaries
Text preservation	utterance logs and source/version marker audit
Channel stress	corruption configuration and accepted interviews
Forged version	attacker configuration, origin checks, final verdicts

B Extended Implications and Boundaries

Nested causal estimands. The comparisons answer different questions. End-to-end communication is endogenous: a memory that changes one speaker’s context can alter downstream evidence. Primary fixed-event replay removes that feedback but conditions on Prov-generated streams and combines all local pipeline differences. Cross-source replay repeats the pipeline contrast on GA text while supplying oracle task versions. Selector-only replay is more causally

specific because it fixes local representation and prompting, but it too assumes correct annotations.

Table 3: Nested comparisons and the claims they support.

Comparison	Held fixed	Supported estimand
End to end	task, model, schedules	Total interface effect; dialogue may mediate
Fixed event	received stream	Conditional pipeline effect on Prov support
Cross source	GA text, task versions	Conditional pipeline effect beyond Prov text
Selector only	candidates, context, interview	Maximum version versus frequency

Four-stage reliability chain. The experiments separate four requirements that a monolithic “memory quality” score would hide. First, the update must *reach* the listener: provenance reaches only 10–19 agents in the primary fixed streams, and the lowest-coverage seed has the smallest complete-pipeline lift. Second, metadata must *survive* communication: default text-only dialogue preserves no parseable source/version marker in 720 utterances, whereas an explicit norm preserves 610. Third, the listener must *resolve* conflicts using that metadata: maximum-version selection gains 9.0 points over mention frequency. Fourth, metadata must be *authenticated*: a forged version 999 hijacks naive Prov, while origin anchoring rejects the tested unauthenticated claim.

Interface fallbacks. Failure of reach should produce unknown rather than an invented current value. Failed preservation should trigger a request for metadata or direct source confirmation. Same-version conflicts should remain unresolved until corroborated, and failed authentication should quarantine the claim rather than merely lower its textual relevance. Such interfaces are useful for auditable coordination but may be undesirable in descriptive social simulations, where selective absorption and provenance loss are behaviors to reproduce.

Table 4: Natural levers do not provide a reliable repair. Brackets are seed-level 95% intervals where reported; values without intervals are exploratory point estimates.

Lever	Conditions (<i>n</i>)	Current HELD (%)
Capability	mini / 5.4 / 5.5 (8/5/5)	18 [12,25] / 34 [25,44] / 30 [21,40]
Connectivity	1 / 2 / 3 meetings (8 each)	20 [13,26] / 18 [12,25] / 14 [10,18]
Persona	thin / rich (5 each)	12 / 14
Authority	base / source (5 paired)	12 [3,21] / 15 [10,21]
Memory motifs	Raw–MemoryBank (8 each)	14–25
Positive control	overwrite (5)	99 [97,100]

Scope and ethical boundary. The harness is a 25-agent, short-horizon simulation over event-planning facts and few model families; it does not cover negotiation, strategic deception, multi-hop planning, or real users. Exploratory long-horizon cells are underpowered. The primary replay conditions on PROV-generated support, while the cross-source and selector tests use oracle task annotations. Controlled adaptations are not official system benchmarks, the semantic audit covers only the 375 core interviews, and the three-seed APM pilot does not establish robustness to forged origins, compromised sources, collusion, replay, or path manipulation.

The study contains no human participants, private records, or demographic inference. Its risks are dual use, anthropomorphic over-interpretation, surveillance through provenance traces, and misplaced trust in credentials. Deployment should minimize retained content, protect and rotate credentials, expose uncertainty and abstention, and undergo stakeholder validation.

C Task, Measures, and Memory Contract

C.1 Registered task state and exposure

The principal repair-drive scenario registers exactly two admissible event states:

$$\begin{aligned} v_0 &= \text{Saturday at the front porch,} \\ v_1 &= \text{Sunday at the community center.} \end{aligned}$$

The environment initializes agents with v_0 and injects v_1 into the designated source after the first round’s encounters. The book-club, carpool, and dues checks preserve the same one-change structure but substitute scenario-specific values. Marker classifiers receive the registered pair for that scenario; they do not search for arbitrary world knowledge.

For agent i , let $E_{qri} = (e_{qri1}, \dots, e_{qrim_{qri}})$ be the events delivered in arrival order. Using the predicates from Section 3.2, current reach and current-marker composition are

$$R_{qri} = \mathbb{I}\left[\sum_j \mathbb{I}[C(e_{qrij})] > 0\right], \quad D_{qri} = \frac{\sum_j \mathbb{I}[C(e_{qrij})]}{\sum_j \mathbb{I}[V(e_{qrij})]}.$$

D_{qri} is undefined when no value-bearing event was received; aggregate summaries report the number of agents with a defined denominator. These diagnostics describe input exposure and do not treat receipt as adoption.

Table 5: Core execution contract.

Item	Registered setting
Society size	25 agents
Transmission horizon	5 rounds unless stated otherwise
Meeting density	2 pairwise meetings per agent per round
Meeting length	3 alternating utterances
Independent unit	schedule seed
Core paired seeds	41–48
Primary generator	gpt-5.4-mini, temperature 0
Model-family check	DeepSeek-V4-Flash
Interview output	one JSON sentence, read-only
Verdicts	current, stale, unknown

C.2 Prompt contract

The conversation system message is:

You are role-playing a person in a small neighborhood. Stay in character and stay consistent with your memory notes – only act on what is currently true and honor any commitments you have made. Reply with ONE short, natural utterance (1–2 sentences). Return ONLY JSON: {"utterance": "..."}.

The conversation user message supplies the speaker persona, listener, topic, retrieved memory notes, optional communication norm, prior utterances in that meeting, and the question “What does [speaker] say next?” The default prompt does not request a source or version; the strong-attribution condition adds that requirement explicitly.

The interview system message is:

Answer the question using ONLY the memory notes, reflecting the CURRENT state. Reply with one short sentence. Return ONLY JSON: {"answer": "..."}.

The interview call cannot modify memory, and no subsequent conversation occurs. Thus HELD is elicited from the final retrieval context rather than self-reported during transmission.

C.3 Memory interfaces

Every observed utterance has an immutable arrival index, round, speaker, listener, and text. Structured conditions may additionally carry an origin, version, and typed value. The raw log is retained even when a memory maintains a compact derived state, allowing the audit to separate what arrived from what the memory accepted.

The GA adaptation appends observed events, produces up to three free-text reflections at round boundaries from the ten most recent events, and retrieves six events plus three reflections for the interview. The PROV pipeline appends the same raw events and maintains a pair (z_i, v_i) initialized to $(0, v_0)$. For each typed claim (z_e, v_e) , it replaces that pair only when $z_e > z_i$. Retrieval prepends the accepted pair and returns four relevance-ranked raw events. This complete-pipeline contrast therefore includes representation, consolidation, selection, retrieval budget, and salience.

The stricter ablation creates a common candidate multiset

$$C_i = \{(a_k, z_k, v_k) : k = 1, \dots, m_i\},$$

where a_k is the arrival index. Both policies use the same store, no-op consolidation, one-record retrieval budget, context string, model call, and verdict function. The frequency policy returns the value with the largest candidate count and breaks ties by latest arrival; the version policy returns the candidate with largest z_k and uses the same tie break.

Table 6: Controlled operations in the selector-only ablation.

Operation	Frequency	Maximum version
Candidate schema	identical	identical
Candidate multiset	identical	identical
Consolidation	no-op	no-op
Retrieval budget	one record	one record
Context template	identical	identical
Interview/scorer	identical	identical
Conflict rule	count, then arrival	version, then arrival

D Replay Construction and Paired Evidence

For each seed, the replay builder reads the accepted round logs, expands every meeting into ordered listener observations, and sorts them by round and arrival index. It checks the reconstructed terminal version and value of each agent against the saved PROV snapshot; a mismatch aborts replay. Across the eight societies, the normalization audit partitions 5,412 received events into 494 structured provenance candidates, 556 unambiguous text-derived candidates, 40 ambiguous mentions excluded from selection, and 4,322 events containing neither registered value. Text-derived versions are analysis annotations based on the registered v_0/v_1 pair, not evidence of natural-language provenance extraction.

all eight seeds but leaves unknown fixed at 23 and converts 18 stale outcomes to current. These distinct error signatures support the nested interpretation in the body: version priority is one verified component, while explicit state representation and retrieval salience contribute additional pipeline-level benefit.

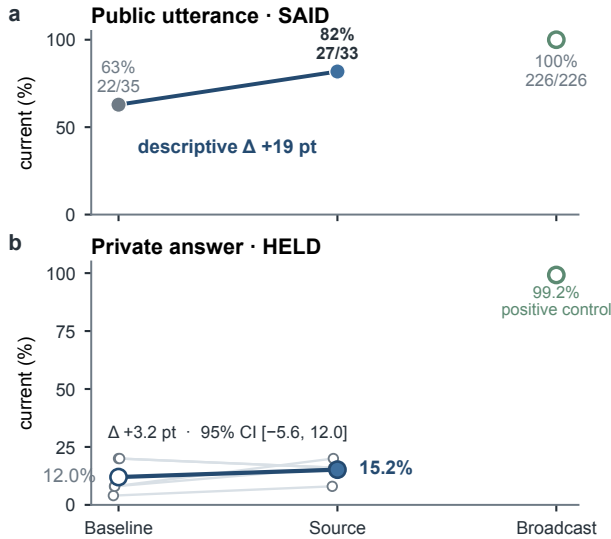


Figure 5: Public uptake, weak retrieval. a, Pooled public current mentions increase under a persistent source. b, Paired private answers remain near baseline. Broadcast is an overwrite positive control; points, intervals, and sample sizes are shown in the panels.

Both complete-pipeline comparisons improve all eight seeds. On PROV-generated streams, the listener change shifts 57 pooled answers to current, primarily by reducing unknown outcomes. On GA-generated streams, it shifts 52 to current but also 22 to stale when only version-0 evidence is available; its net current-minus-stale interval crosses zero. The selector-only comparison improves

Table 7: Per-seed current HELD rates for three paired replay experiments. Each seed cell is a percentage over 25 agents. The pooled rows report verdict counts over 200 agents per policy; all 400 selector interviews matched the selected symbolic state.

Replay	Policy	41	42	43	44	45	46	47	48	Mean
Complete pipeline	GA	36	36	24	28	28	36	36	32	32.0
	PROV	72	64	64	60	40	60	64	60	60.5
	Difference	36	28	40	32	12	24	28	28	28.5
GA-source pipeline	GA	32	20	16	16	8	12	12	4	15.0
	PROV	56	52	60	48	40	20	28	24	41.0
	Difference	24	32	44	32	32	8	16	20	26.0
Selector only	Mention frequency	72	68	80	68	52	44	60	56	62.5
	Maximum version	76	76	84	76	56	64	72	68	71.5
	Difference	4	8	4	8	4	20	12	12	9.0

Replay	Policy	Current	Stale	Unknown
Complete pipeline	GA / PROV	64 / 121	15 / 10	121 / 69
GA-source pipeline	GA / PROV	30 / 82	16 / 38	154 / 80
Selector only	Frequency / version	125 / 143	52 / 34	23 / 23

E Robustness and Security Boundaries

Scenario, fact-type, topology, and model-family results are reported once, in Table 1. The remaining diagnostics test whether provenance survives the communication channel and whether a larger version number can be forged.

Table 8: Channel and security boundary diagnostics.

Test	Condition	Current	Stale	Unknown
Text only	Default	16	0	59
	Strong attribution	75	0	0
Channel stress	Metadata drop	93/100/93/26	–	–
	Value garbling	93/74/42/24	–	–
Forged version	Naive PROV	25	43	7
	APM	48	0	27

Text-only and forged-version rows give pooled current/stale/unknown counts. Stress-test current counts correspond to $p = 0.0/0.3/0.6/0.9$, each out of 100. Default and strong-attribution dialogue preserve 0/720 and 610/720 marked utterances.

Default text-only dialogue preserves no parseable source–version marker, and most agents abstain. A strong attribution norm preserves 610 of 720 markers and provides an upper bound under protocolized communication. Metadata drop is partly masked by redundant paths, whereas value garbling degrades current answers more steadily because a high version can carry the stale value. These finite-run cells are diagnostics rather than monotonicity estimates.

Naive PROV treats a larger integer as authoritative. The forged-version pilot therefore gives one relay an unauthenticated stale claim labeled version 999. The origin-anchored extension APM accepts a versioned update only when its origin matches the registered authority and retains the relay path for audit. It does not claim cryptographic security: the experiment does not cover a compromised authority, forged origin credentials, collusion, replay, or path manipulation.

In the eight-seed sparse comparison, APM yields 40.5% current versus 21.5% for GA, with 0/200 versus 30/200 stale answers. The

safety-relevant result is the absence of accepted stale answers under the tested attacker, not a general claim of attack resistance. Abstention rises because rejecting unauthenticated claims deliberately sacrifices coverage.